Floor Area and Regional Differences in New Home Prices

STA302H1 Fall 2026 | Group 62 | Part 1 Research Proposal

Contributions

Group 62

Member name	Contribution to this proposal
Haiwen Yi	Research design, statistical analysis, and report writing.
Olivia Qi	Revision-result Interpretation, details, structure and format review
Matthew Frieri	Revised proposal against rubric, bringing sections up to par with criterion
Annie Xu	_____

Introduction

Among new detached, non-condominium homes sold in the United States in 2025, how is sale price (including land, in thousands of dollars) associated with finished floor area (in hundreds of square feet), and does this association differ across the nine Census divisions (regional groupings of states) after adjustment for lot size, room counts, amenities, construction features and sale timing?

Prior evidence gives two reasons to question a uniform area–price relationship. In a meta-regression of U.S. single-family housing studies, Sirmans et al. (2006) found geographic differences in floor-area coefficients and reported that model size affected those coefficients. This motivates comparing regions using the same adjustment set. A possible economic mechanism is variation in building costs: across U.S. housing markets, Gyourko and Saiz (2006) found substantial cost differences associated with wages, unionization, terrain and regulation. Because our response includes both structure and land, however, a regional area coefficient cannot be interpreted as construction cost. Heterogeneity also occurs within markets: using over 136,000 single-family sales in Jacksonville, Florida, during 1990–2006, Zietz et al. (2008) found larger implicit prices for floor area and lot size at higher price quantiles. Thus, geographic and price-segment heterogeneity are distinct; their quantile results do not establish differences between regional mean slopes.

Our contribution is a common-data comparison for 2025 new detached homes: we estimate regional conditional mean slopes under one population definition, year and specification, rather than compare coefficients assembled from different studies. Adjusting for lot size and room counts makes the area coefficient describe differences in interior space at otherwise fixed recorded characteristics. The analysis tests regional variation but cannot identify whether costs, preferences or unmeasured location explain it.

Regional averages and pairwise correlations cannot provide these conditional comparisons. Multiple linear regression includes floor area, division and their interaction alongside the adjustment variables. Nine estimated slopes and confidence intervals quantify price differences associated with 100 additional square feet; a joint test of the eight interaction coefficients assesses whether the slopes differ. Builders and appraisers could use these estimates to judge whether one national size adjustment obscures regional differences, as background for local market assessment. They are not individual appraisals or causal returns to expansion. The linear form requires diagnostic checking, and unweighted complete-case analysis limits population generalization.

Data description

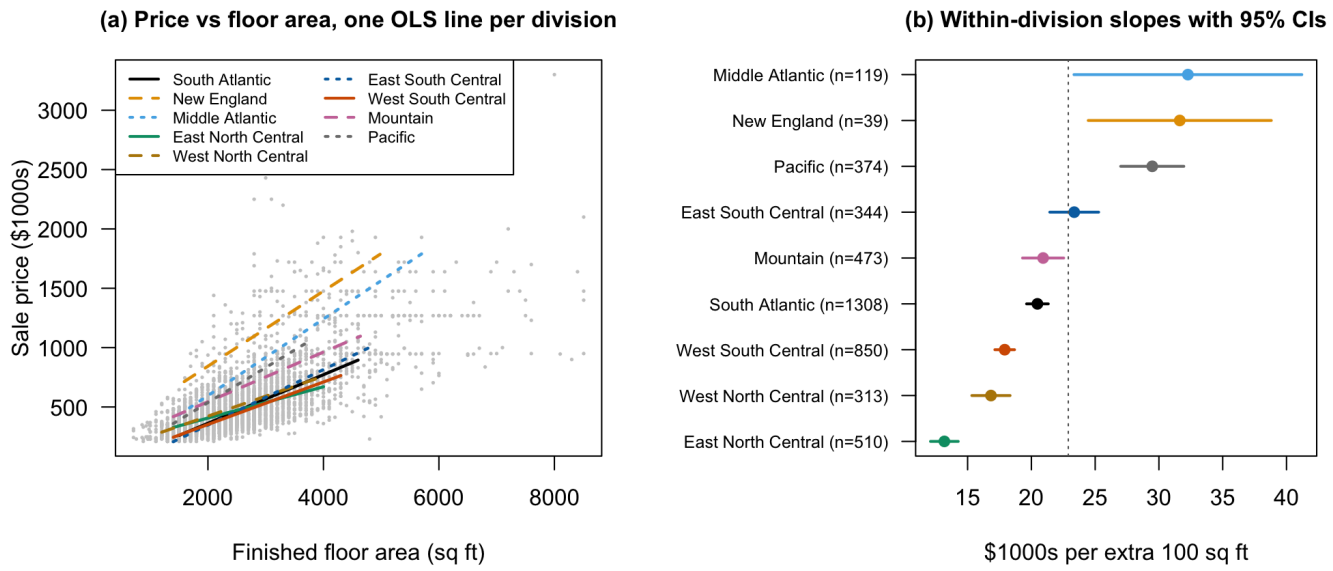
The 2025 Survey of Construction (SOC) microdata support national/regional construction/sales statistics through permit sampling, non-permit canvassing and builder/owner interviews (U.S. Census Bureau, n.d.-a, 2026). Of 25,246 records, 7,382 match our target population. Sequential exclusions for unreported price (320), area/lot (747) and categorical predictors (130) leave 6,185 complete cases (4,330 training; 1,855 test); excluding 16.2% may bias estimates. Approximately continuous price (US$210k–3.3M; Table 1) supports conditional-mean regression as a starting model; shared builders may compromise independence. Area-by-division interaction tests adjusted geographic slope differences (Sirmans et al., 2006; Figure 1); Census divisions remain unchanged.

Table 1. *Training sample summaries.*

Variable	Summary
Price ($1,000)	527 (307); median: 433; range: 210–3,300
Area (100sq ft)	25.4 (9.9)
Lot (1,000sq ft)	14.1 (16.6)
Sale month	6.3 (3.4)
Division	South Atlantic: 1,308; New England: 39; Middle Atlantic: 119; East North Central: 510; West North Central: 313; East South Central: 344; West South Central: 850; Mountain: 473; Pacific: 374
Stories	1: 2,120; 2: 2,116; 3+: 94
Bedrooms	≤2: 154; 3: 1,780; 4: 1,777; 5+: 619
Full bathrooms	≤1: 46; 2: 2,664; 3: 1,203; 4+: 417
Half bathrooms	0: 2,373; 1: 1,897; 2+: 60
Garage	1-car: 158; 2-car: 3,124; 3+-car: 919; other: 129
Foundation	Basement: 992; crawl space: 403; slab: 2,926; other: 9
Fireplaces	0: 2,781; 1: 1,383; 2+: 166
HOA	No: 1,109; yes: 3,221
Age restriction	No: 4,216; yes: 114
Wall	Brick: 897; wood: 169; stucco: 906; vinyl: 1,362; stone: 56; fiber cement: 926; other: 14
Heating	Electric: 2,147; gas: 2,083; propane: 52; other: 48
Sewer	Public: 3,893; septic/other: 437
Sale stage	Completed: 2,452; under construction: 1,344; not started: 534

Note. Numerical: mean (standard deviation); price also median/range. Categories: counts, reference levels first. Price, area and lot are right-skewed; extremes are disclosure-coded (U.S. Census Bureau, n.d.-b). New England's 39 observations limit precision.

Figure 1. *Unadjusted regional price–area relationships.*



Ethics discussion

The first ethics module asks whether data are collected or simulated, whether metadata and original sources are documented, and whether third-party use supports trustworthiness. SOC records come from a documented Census survey, not simulated examples; the published definitions, sampling information and disclosure flags provide traceable metadata. On third-party use, SOC data underlie official federal economic indicators and are used by HUD, the Bureau of Economic Analysis, the National Association of Home Builders and Harvard's Joint Center for Housing Studies. Government origin supports provenance but does not remove sampling, reporting or measurement error.

For privacy, the public file suppresses detailed geographic information, removes addresses and other identifiers, and codes extreme values. We will not attempt re-identification or publish individual-house appraisals. Regarding consent, public statistical release is distinct from participants consenting to arbitrary new uses: we limit our use to aggregate housing research compatible with the collection purpose and do not claim to have independently audited every respondent's consent. Regarding ownership, we acknowledge the Census Bureau and HUD and retain source attribution. The estimates describe conditional associations; regional labels must not be used to stereotype households or imply causal effects of adding floor space.

Preliminary results

Unweighted OLS includes 17 predictors and the area–division interaction (52 coefficients; n=4,330). Training R^2=0.713 (adjusted R^2=0.710), the interaction adds 0.020. A conventional partial F-test of the eight area-division interaction terms gives $F(8,4278)=37.27$, $p<0.001$, indicating regional slope differences if the usual OLS assumptions hold.

Holding other predictors fixed (Table 2), 100 additional square feet is associated with a fitted mean price increase of $11,834 in South Atlantic and $24,557 in Middle Atlantic. The estimated difference between these slopes is $12,723 per 100 square feet (95% CI: $10,359–$15,087). New England's slope estimate is less precise because it is based on only 39 observations.

Condition 1 (conditional mean response): Figure 2A broadly follows the identity line centrally, but sparse upper-range departures limit certainty. Condition 2 (conditional mean predictor) requires linear numerical-predictor conditional means; lot–area curvature raises concern (Figure 2B), despite flatter month relationships (2C–D). Figure 3A–B shows curvature, fanning and rising scale–location, suggesting mean/variance problems. Condition concerns prevent isolating violations. The area smoother bends at high values (Figure 3D); lot/month smoothers remain near zero (Figure 3E–F). The Q–Q plot (Figure 3C) shows heavy tails, with greater right-tail departure; 36 standardized residuals exceed ±4 (Figure 4F). Group spreads differ (Figures 4A–E, 5); regional SDs vary fivefold. Independence remains uncertain without builder/subdivision identifiers. Inference remains provisional; no corrections applied.

Regional slope differences agree with Sirmans et al. (2006). Construction costs offer a possible but untested explanation (Gyourko & Saiz, 2006). Our analysis cannot establish that costs cause the observed differences.

Table 2. *Adjusted floor area slopes by Census division ($1,000 per 100 sq ft; South Atlantic reference).*

Division	n	Slope	SE	95% CI	Difference from South Atlantic (SE)
South Atlantic (reference)	1,308	11.83	0.63	(10.60, 13.07)	—
New England	39	19.26	2.15	(15.05, 23.47)	7.43 (2.17)
Middle Atlantic	119	24.56	1.14	(22.32, 26.79)	12.72 (1.21)
East North Central	510	6.85	0.85	(5.19, 8.51)	−4.98 (0.91)
West North Central	313	10.57	1.15	(8.32, 12.82)	−1.26 (1.20)
East South Central	344	13.78	0.97	(11.88, 15.68)	1.95 (1.00)
West South Central	850	9.27	0.79	(7.71, 10.82)	−2.57 (0.82)
Mountain	473	14.60	0.80	(13.03, 16.17)	2.77 (0.87)
Pacific	374	21.70	1.14	(19.47, 23.92)	9.86 (1.17)

Figure 2. *Conditional-mean response and predictor checks.*

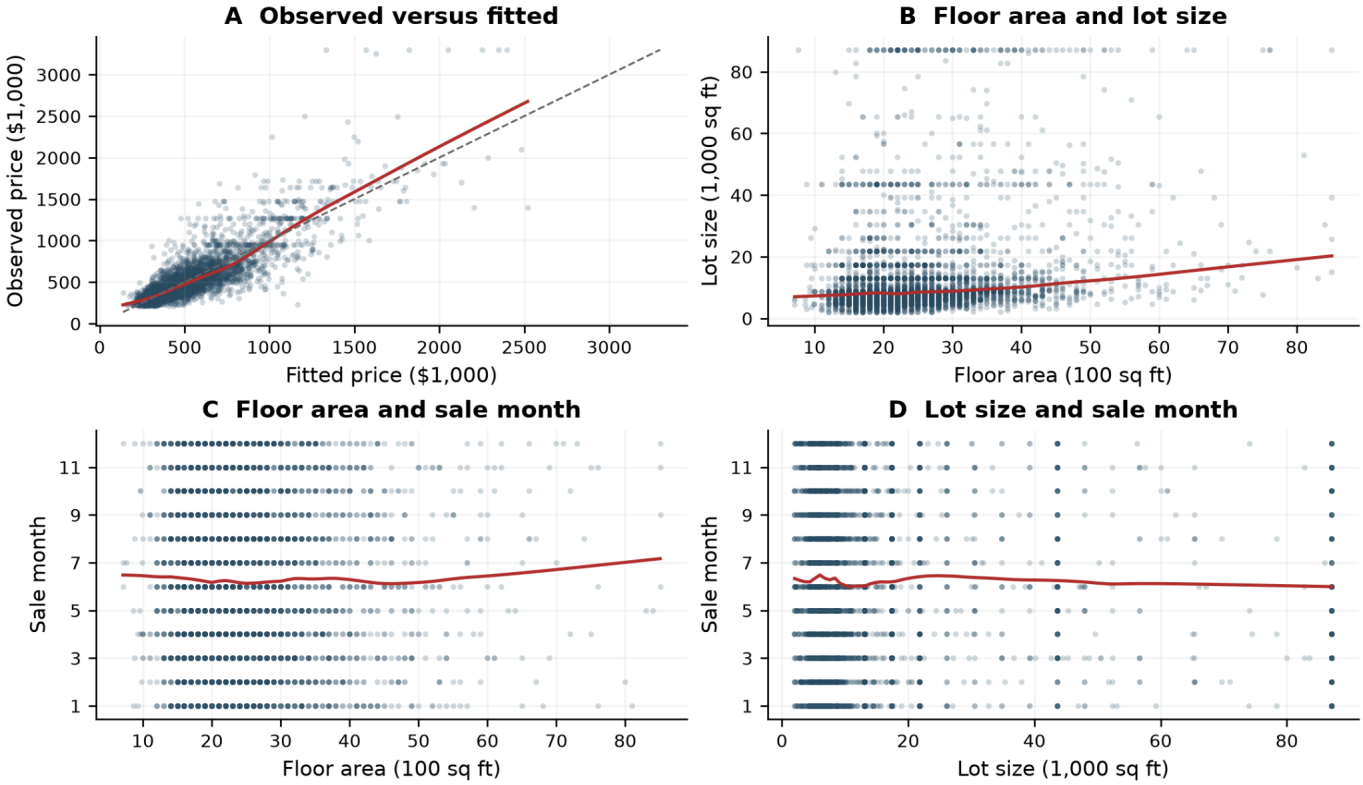


Figure 3. *Residual diagnostics for the preliminary regression model.*

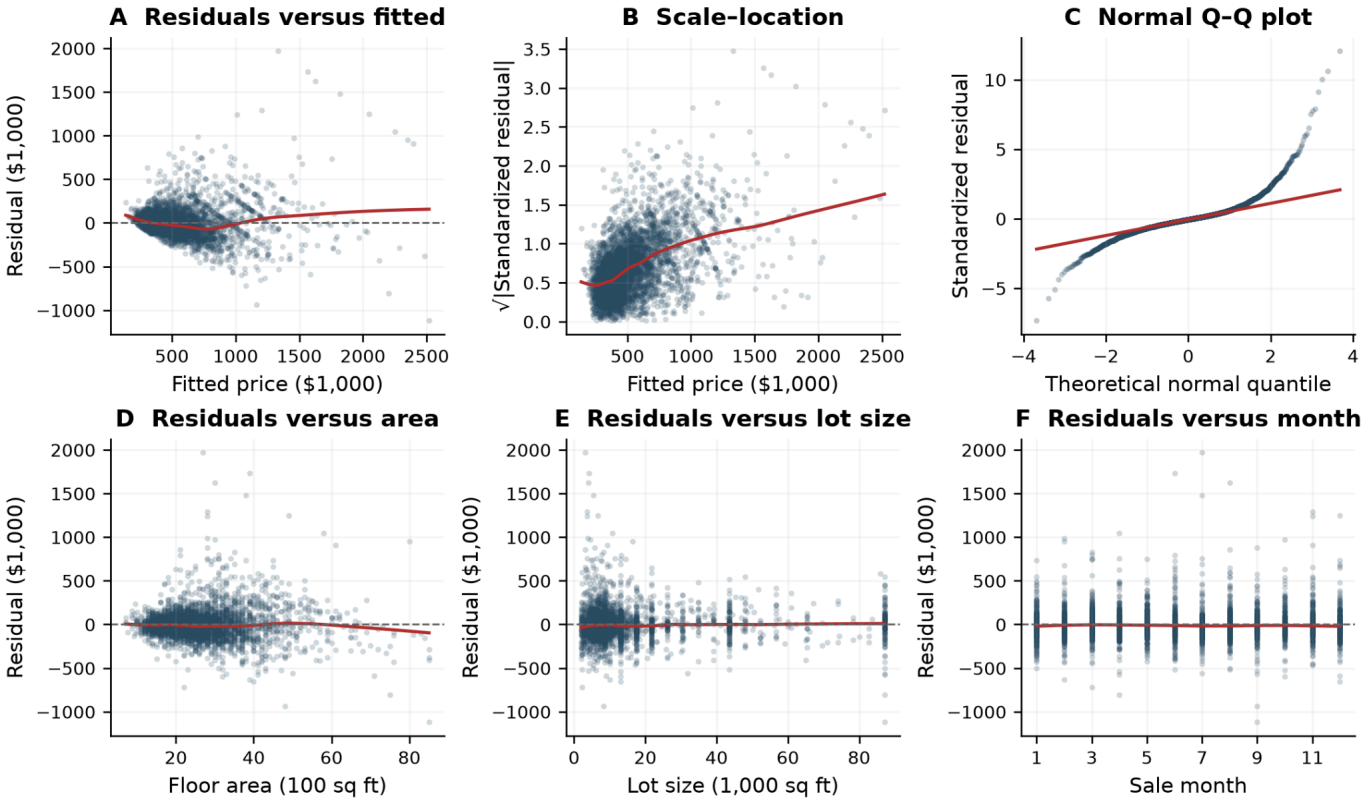


Figure 4. *Selected categorical residual distributions and standardized-residual histogram.* SD ratios compare largest/smallest within-group SDs; boxplots omit outlier points.

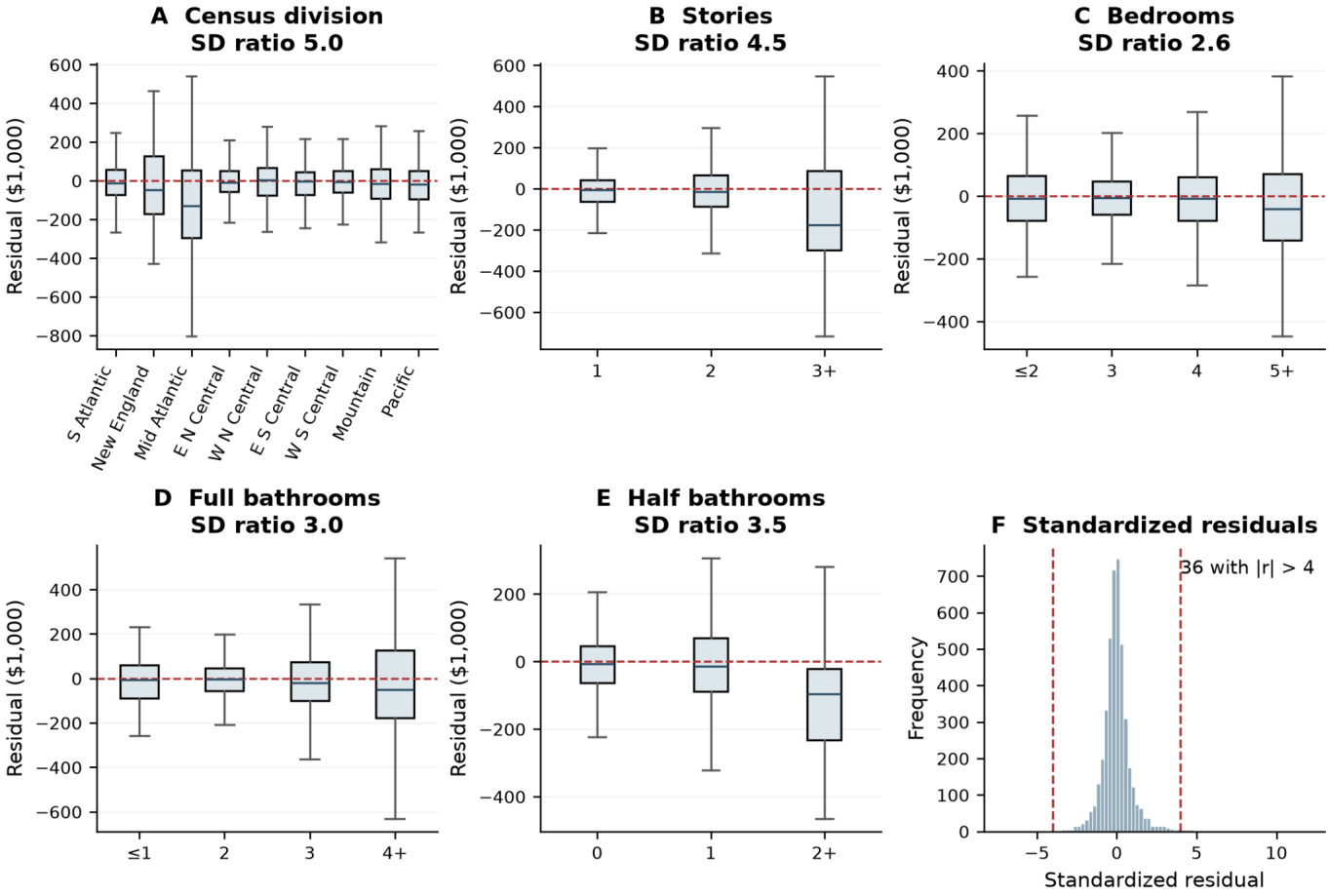
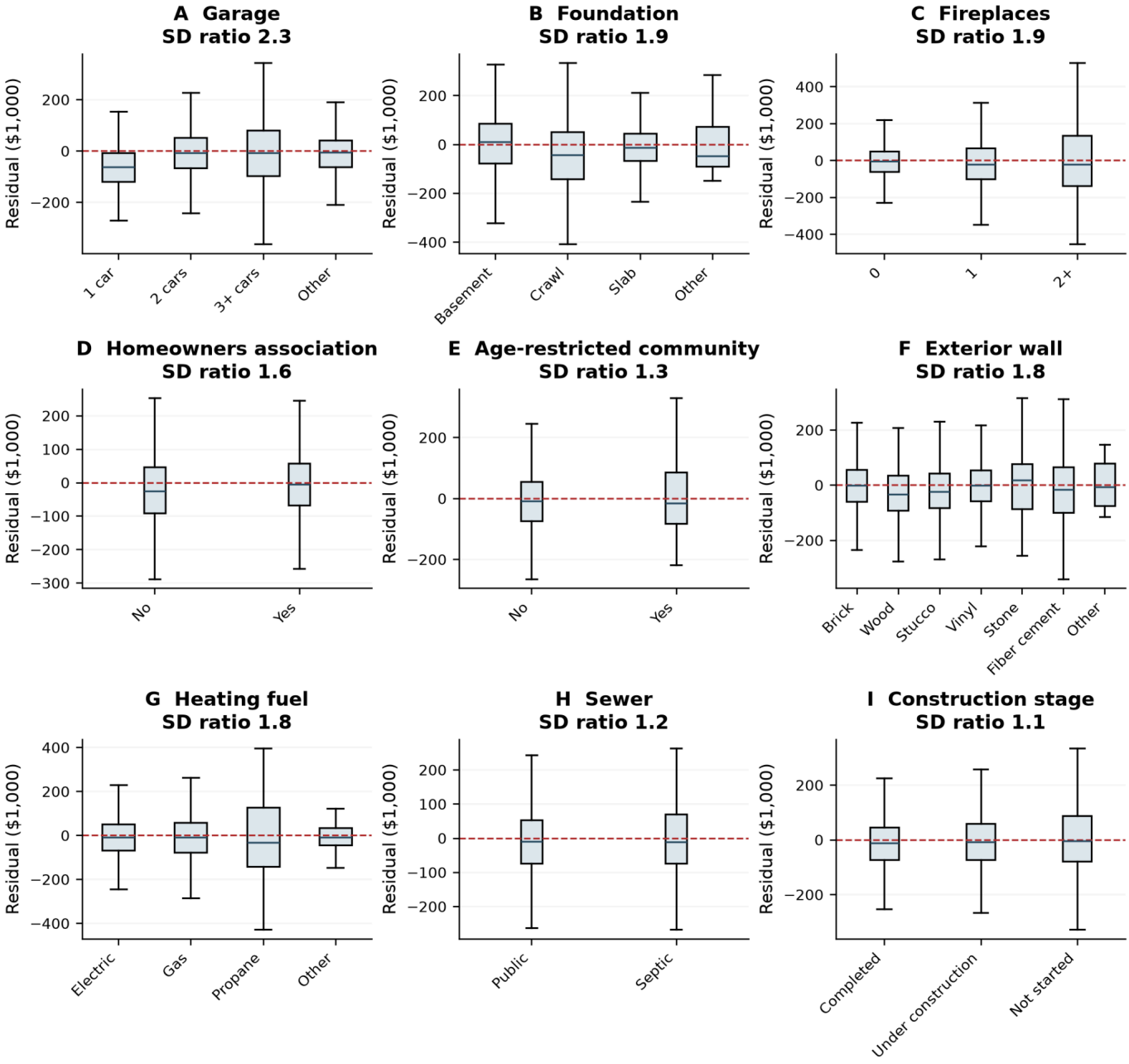


Figure 5. *Additional categorical residual distributions; outlier points hidden.*



Plan

Retain floor area, division, their interaction and lot size for the research question and hedonic context (Rosen, 1974). Use grouped backward BIC selection, AIC sensitivity checks and appropriate nested F-tests, preserving interaction hierarchy.

First assess Box–Cox price transformations, including log price, for the fanning and heavy tails. If curvature remains, compare log-area or quadratic-area terms. Recheck Conditions 1–2 and residual diagnostics after every model change; transformation is not guaranteed to resolve misspecification.

After improving the mean structure, use HC3 inference if unequal variance persists; consider weighted least squares only with a defensible variance model. Assess influential cases and survey-weight sensitivity; remove observations only for documented errors. Unidentified clustering remains a limitation that HC3 cannot fix.

Select transformations using training diagnostics, then freeze the model before one held-out test evaluation.

Table 3. *Team schedule.*

Date	**Deliverable**	**Lead**
Nov 5	Submit Parts 0–1.	All
Before final submission	Hold at least two TA/professor consultations for rubric-based feedback; document suggestions, revise the project and verify improvements.	Coordinator: TBD
Nov 19	Compare models; assess influential cases.	Haiwen Yi
Nov 22	Complete robustness checks; freeze model; evaluate test set.	Haiwen Yi
Nov 27	Draft poster content, figures and layout.	Olivia Qi
Nov 27	Complete presentation script and visual design.	TBD
Nov 30	Complete poster, rehearsal and recording.	All
Dec 1	Check rubric, numbers, citations and recording; return corrections.	TBD
Dec 2	Resolve feedback; finalize files and individual reflections.	All
Dec 3	Submit Parts 2–3; verify receipts.	All

Internal targets except submissions (23:59 Toronto time). Consultations subject to availability; TBD roles await volunteers.

Bibliography

Gyourko, J., & Saiz, A. (2006). Construction costs and the supply of housing structure. *Journal of Regional Science, 46*(4), 661–680. https://doi.org/10.1111/j.1467-9787.2006.00472.x

Rosen, S. (1974). Hedonic prices and implicit markets: Product differentiation in pure competition. *Journal of Political Economy, 82*(1), 34–55. https://doi.org/10.1086/260169

Sirmans, G. S., MacDonald, L., Macpherson, D. A., & Zietz, E. N. (2006). The value of housing characteristics: A meta analysis. *The Journal of Real Estate Finance and Economics, 33*(3), 215–240. https://doi.org/10.1007/s11146-006-9983-5

U.S. Census Bureau. (n.d.-a). *Survey of Construction methodology.* https://www.census.gov/construction/soc/methodology.html

U.S. Census Bureau. (n.d.-b). *Survey of Construction microdata files: General information and variable descriptions* [Documentation]. https://www.census.gov/construction/chars/pdf/socmicro_info.pdf

U.S. Census Bureau. (2026). *Survey of Construction (SOC) microdata: 2025 annual file* [Data set]. https://www.census.gov/construction/chars/microdata.html

Zietz, J., Sirmans, G. S., & Smersh, G. (2008). The impact of inflation on home prices and the valuation of housing characteristics across the price distribution. *Journal of Housing Research, 17*(2), 119–137. https://doi.org/10.1080/10835547.2008.12091992